

International Institute of Information Technology, Hyderabad

MA4.101 Real Analysis (Monsoon 2026)

Assignment 1 — Grading Rubrics

Total Marks: 75

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Question	1	2	3	4	5	6	7	8	Total
Marks	15	10	8	8	10	8	6	10	75

Grading Rubric for Question 1

Total Marks: 15

General Marking Principles (apply to every part)

- **Accept any correct proof.** The model solution is one route, not the only one. A different but valid induction, case split or argument earns full marks.
- **Earlier parts may be used freely,** even if the student did not prove them (or proved them incorrectly). Do not re-penalise the same mistake in later parts (follow-through marking).
- **Standard facts about $+$, $\times$ and $\leq$ on $\mathbb{N}$** (associativity, commutativity, cancellation, distributivity, trichotomy, $(n+k)++ = n+(k++)$, multiplication preserves $\leq$) may be used without proof or citation.
- **Missing citations are not penalised.** If a step is correct but the student does not name the clause or part used, give full credit.
- **Minor slips** (notation, a mislabelled variable, a missing " $0 \ominus 0 = 0$ " line, a typo that does not change the logic) cost nothing.
- **Ordinary subtraction.** If a student uses $m - n$ where only $\ominus$ is available, deduct at most 0.5 in that part, and only if the argument genuinely depends on it.
- **Partial credit is encouraged.** Award marks for a correct setup (e.g. naming the right induction variable, stating the right cases) even if the proof is not finished. Where a sub-criterion is half done, round up to the nearest 0.5.
- **Not required:** checking that pred and $\ominus$ are well defined via the recursion theorem (mentioned in the Rationale box). No marks are lost if this is omitted.

(a) $0 \ominus n = 0$ for all n — 1 Mark

Criteria	Description	Marks
Base Case ($n = 0$)	States $0 \ominus 0 = 0$ from the first clause of $\ominus$. Accept even if just written down as obvious.	0.5
Inductive Step	Uses $0 \ominus n++ = \text{pred}(0 \ominus n) = \text{pred}(0) = 0$. Award in full if the computation is right, even if the inductive hypothesis is not stated explicitly.	0.5

(b) $m++ \ominus n++ = m \ominus n$ — 1 Mark

Criteria	Description	Marks
Base Case ($n = 0$)	Shows $m++ \ominus 0++ = \text{pred}(m++) = m = m \ominus 0$.	0.5
Inductive Step	Uses the hypothesis inside pred : $m++ \ominus n++++ = \text{pred}(m++ \ominus n++) = \text{pred}(m \ominus n) = m \ominus n++$. Award 0.5 if the student inducts on m instead but the argument is still valid.	0.5

(c) $(m \ominus n) \ominus k = m \ominus (n + k)$ — 2 Marks

Criteria	Description	Marks
Setup and Base Case	Chooses a sensible induction variable (k is natural; any variable that works is fine) and checks $k = 0$: both sides equal $m \ominus n$.	0.5
Inductive Step: definition	Rewrites the left side as $\text{pred}((m \ominus n) \ominus k)$ and applies the hypothesis to get $\text{pred}(m \ominus (n + k))$.	1
Inductive Step: conclusion	Recognises $\text{pred}(m \ominus (n + k)) = m \ominus (n + k)++$ and uses $(n + k)++ = n + k++$. Do not deduct if the last addition fact is used silently.	0.5

(d) $(m + n) \ominus n = m$ — 1 Mark

Criteria	Description	Marks
Base Case ($n = 0$)	$(m + 0) \ominus 0 = m$.	0.5
Inductive Step	Writes $m + n++ = (m + n)++$ and applies part (b), then the hypothesis. Full credit if (b) is used without being named.	0.5

(e) $m \ominus n = 0 \Leftrightarrow m \leq n$ — 2 Marks

Criteria	Description	Marks
Lemma $m \ominus m = 0$	Any justification (e.g. part (d) with m replaced by 0, or a direct induction). Give the mark if the fact is used correctly with a one-line reason.	0.5
$(\Leftarrow)$ direction	Writes $n = m + k$ and uses (c) and (a): $m \ominus (m + k) = (m \ominus m) \ominus k = 0 \ominus k = 0$.	0.5
$(\Rightarrow)$ direction	Contrapositive or contradiction: if $m \not\leq n$ then $m = n + k$ with $k \neq 0$ (0.5), and (d) gives $m \ominus n = k \neq 0$ (0.5). A direct proof by induction is equally acceptable.	1

(f) $(m \ominus n) + n = m \Leftrightarrow n \leq m$ — 2 Marks

Criteria	Description	Marks
$(\Leftarrow)$ direction	Writes $m = n + j$, uses (d) to get $m \ominus n = j$, then adds n back.	1
$(\Rightarrow)$ direction	Observes that $m = n + (m \ominus n)$ exhibits m as n plus a natural number, which is the definition of $n \leq m$. This is short: a one-line correct observation earns the full mark.	1

(g) $\max(m, n) = m + (n \ominus m)$, and the symmetric identity — 2 Marks

Criteria	Description	Marks
Case $n \leq m$	Uses (e) to get $n \ominus m = 0$, so $m + 0 = m = \max(m, n)$.	0.5
Case $n \not\leq m$	Uses trichotomy to write $n = m + k$ and (d) (or (f)) to get $n \ominus m = k$, so $m + k = n = \max(m, n)$.	0.5
Symmetry of max	Any argument that $\max(m, n) = \max(n, m)$. Be lenient: a brief reason such as “both equal the larger of the two” earns the mark; if symmetry is used silently, still award this mark; deduct only if the reasoning given is actually wrong.	0.5
Deduction	Applies the identity with m, n swapped and chains the equalities to get $m + (n \ominus m) = n + (m \ominus n)$. Also accept a direct proof of this identity by cases without max.	0.5

(h) $k \times (m \ominus n) = (k \times m) \ominus (k \times n)$ — 2 Marks

Criteria	Description	Marks
Case Split	Splits into $n \leq m$ and $m \leq n$ (via trichotomy), or sets up another valid approach (e.g. induction).	0.5
Case $n \leq m$	Writes $m = n + j$, uses distributivity $km = kn + kj$ and (d) twice to get $km \ominus kn = kj = k(m \ominus n)$.	1
Case $m \leq n$	Both sides are 0 by (e): $m \ominus n = 0$ and $km \leq kn$. “Multiplication preserves order” may be used without proof. The case $k = 0$ need not be treated separately.	0.5

(i) Cancellation for $\ominus$ and counterexamples — 2 Marks

Criteria	Description	Marks
Main Implication	Uses (f) on both m and n : $m = (m \ominus k) + k = (n \ominus k) + k = n$. Alternatively cancel via (d) after writing $m = k + a$, $n = k + b$.	1
Counterexample	Gives a concrete example where one hypothesis fails and the conclusion fails, e.g. $k = 1$, $m = 1$, $n = 0$ (then $m \ominus k = n \ominus k = 0$ but $m \neq n$). Award 0.5 for a valid example with the computation shown (checking $k \not\leq n$ need not be proved from the axioms; “ $1 > 0$ ” is fine).	0.5
Both Hypotheses	Addresses the other hypothesis, either by a second example or by noting the statement is symmetric in m and n . A remark that swapping m and n handles it earns the full 0.5.	0.5

Mark Summary

Part	a	b	c	d	e	f	g	h	i	Total
Marks	1	1	2	1	2	2	2	2	2	15

Grading Rubric for Question 2

Total Marks: 10

General Marking Principles (apply to every part)

- **Accept any correct proof.** The model solution is one route, not the only one.
- **Earlier parts may be used freely** in later parts, even if not proved (or proved incorrectly). Do not re-penalise the same mistake (follow-through marking).
- **Standard facts about $+$, $\leq$ on $\mathbb{N}$** may be used without proof. The *Peano axioms* that are used (0 is not a successor; successor is injective; induction) must be visibly *used*; a missing axiom *number* costs nothing if the use is clear.
- **Division, parity and “ $\lfloor n/2 \rfloor$ ” are not available** (assignment guideline 1). Informal remarks such as “ $h(n)$ is the lower half” are fine as intuition but cannot replace a construction.
- **Minor slips** (notation, typos, a mislabelled variable) cost nothing.
- **Partial credit is encouraged.** Where a sub-criterion is half done, round up to the nearest 0.5.

(a) Three forms, uniqueness of m , and why it is needed — 2 Marks

Criteria	Description	Marks
Exhaustiveness	Every n is 0, $0++$ or $m++++$, by induction on n (cases $n = 0$, $n = 0++$, $n = m++++$ for the step). A correct case analysis on “ n is 0 or a successor” applied twice also earns the mark.	0.5
Mutual exclusivity	0 is not a successor (Axiom 2.3); $0++ = m++++$ forces $0 = m++$ by injectivity (Axiom 2.4), again contradicting 2.3.	0.5
Uniqueness of m	$m++++ = m'++++ \Rightarrow m = m'$ by two applications of injectivity.	0.5
Why needed	Explains that a definition must assign <i>every</i> n a value (exhaustiveness) and <i>never two</i> (exclusivity + uniqueness of m). Award the full 0.5 for a correct explanation of either half with the other clearly implied.	0.5

(b) Why Eq. (1) is not an instance of the recursion theorem — 2 Marks

Criteria	Description	Marks
Shape of the theorem	States (in words or symbols) what the theorem gives: $a_0 = c$, $a_{n++} = f_n(a_n)$, i.e. one initial value and the value at $n++$ determined from the value at n .	0.5
Two initial values	Notes that Eq. (1) prescribes two starting values, at 0 and at $0++$.	0.5
Two-step dependence	The decisive point: the third clause gives $h(n++++)$ from $h(n)$, two steps back, and says nothing about $h(n++)$ in terms of $h(n)$; so no family (f_n) is supplied. Award 0.5 if only “it goes back two steps” is said without explaining why that breaks the theorem’s shape.	1

(c) Existence of h — 4 Marks

Two routes are accepted. Mark the student's answer against whichever route they follow; do not mix the tables.

Both routes earn full marks if correctly done. The Rationale box for Q2 describes recursing on pairs in $\mathbb{N} \times \mathbb{N}$, while the model solution recurses into $\mathbb{N}$ with an index-dependent step rule. Since the assignment itself points to both, neither route is penalised. A Route B answer may cite a recursion theorem into an arbitrary set without proving it.

Route A: recursion into $\mathbb{N}$ with an index-dependent step (model solution).

Criteria	Description	Marks
Parity indicator	Constructs, <i>by the recursion theorem</i> , a sequence (p_n) with $p_0 = 0$, $p_{n++} = \varphi(p_n)$ (or any equivalent device that tells the step rule whether to increase).	1
Properties of p_n	Shows $p_n \in \{0, 1\}$ for all n and that p_n alternates (P1, P2). Accept a one-line induction.	0.5
Step rule	Defines $f_n(y) = y$ if $p_n = 0$, $f_n(y) = y++$ otherwise, and notes each f_n is a function $\mathbb{N} \rightarrow \mathbb{N}$.	1
Recovering h	Sets $h(n) := a_n$ for the sequence the theorem returns (with $c = 0$).	0.5
Verification	Checks $h(0) = 0$, $h(0++) = 0$, and $h(n+++++) = h(n)++$ using both cases of p_n . Award 0.5 if the third clause is checked in only one case.	1

Route B: recursion on pairs.

Criteria	Description	Marks
Target set	States that the recursion is into $\mathbb{N} \times \mathbb{N}$ and that this is a set (or cites a form of the recursion theorem valid for an arbitrary set).	0.5
Initial value	$p_0 = (0, 0)$, encoding $(h(0), h(0++))$.	0.5
Step rule	$F(x, y) := (y, x++)$, with $p_{n++} = F(p_n)$. Any rule that carries one step of history and is correct earns the mark.	1
Recovering h	Sets $h(n) :=$ first coordinate of p_n , and observes the function from the theorem is <i>not</i> h itself.	0.5
Verification	Shows by induction that $p_n = (h(n), h(n+++))$ (or equivalent), then checks all three clauses of Eq. (1). Award 1 if the clauses are checked but the invariant is used silently.	1.5

Caps for structurally flawed answers in (c) (apply the lowest applicable cap, then award within it)

Answer type	Max
Simply asserts “ h exists by the recursion theorem” with Eq. (1) as the rule (contradicts part (b)).	0
Defines h by an explicit formula using division or $\lfloor n/2 \rfloor$, with no construction.	0.5
Uses “if n is even / odd” in the step rule with parity never defined, but the rest of Route A is correct.	2.5
Defines h correctly but never verifies Eq. (1).	3

(d) Uniqueness of h — 2 Marks

Criteria	Description	Marks
Strengthened statement	Recognises that plain induction on $h(n) = h'(n)$ does not close, and inducts on $Q(n)$: $h(n) = h'(n)$ and $h(n++) = h'(n++)$. Strong induction is equally acceptable.	0.5
Base case	$Q(0)$ from the first two clauses of Eq. (1).	0.5
Inductive step	First half of $Q(n++)$ is the second half of $Q(n)$; second half from the third clause: $h(n+++++) = h(n)++ = h'(n)++ = h'(n+++++)$.	1

Alternative: deducing uniqueness from the uniqueness clause of the recursion theorem earns full marks *only if* the student shows that every h satisfying Eq. (1) yields a sequence obeying the recursion used in (c) (e.g. $n \mapsto (h(n), h(n++))$ satisfies $p_{n++} = F(p_n)$). A bare “unique by the recursion theorem”: 0.5.

Mark Summary

Part	a	b	c	d	Total
Marks	2	2	4	2	10

Grading Rubric for Question 3

Total Marks: 8

Criteria	Description	Marks Available
Base Case ($k = 0$)	Correctly verifying the base case for $k = 0$ (where $n = 1$), yielding the trivial inequality $a_1 \geq a_1$.	1
Two-Variable Lemma	Formulating and adequately proving the foundational lemma $(S + T)^2 \geq 4ST$ for natural numbers (e.g., substituting $S = T + u$). Penalty: Deduct 1 mark if the proof relies on subtraction (e.g., starting from $(S - T)^2 \geq 0$), as the customary proof is unavailable because $S - T$ need not lie in $\mathbb{N}$.	2 (1 if subtraction is used)
Inductive Hypothesis	Explicitly stating the assumption that the inequality holds for $n = 2^k$.	0.5
Inductive Step Setup	Correctly splitting the sum of $2n$ terms into two halves to utilize the lemma: $S := a_1 + \cdots + a_n$ and $T := a_{n+1} + \cdots + a_{2n}$.	1
Lemma Application	Raising the foundational lemma inequality to the power of n (which preserves the inequality) to establish $(S + T)^{2n} \geq 4^n S^n T^n$.	1
Hypothesis Application	Correctly applying the induction hypothesis to bound both $S^n \geq n^n \prod_{i=1}^n a_i$ and $T^n \geq n^n \prod_{i=n+1}^{2n} a_i$.	1
Final Simplification	Successfully multiplying the bounded inequalities and algebraically simplifying the coefficient $4^n n^n n^n = (2n)^{2n}$ to complete the proof for $2n = 2^{k+1}$.	1.5

Grading Rubric for Question 4

Total Marks: 8

Question 4 [8 Marks]. Suppose $a_1 \leq a_2 \leq \dots \leq a_n$ and $b_1 \leq b_2 \leq \dots \leq b_n$ are natural numbers. Prove that

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right).$$

Solution. *Pairwise lemma.* For all indices i, j ,

$$a_i b_j + a_j b_i \geq a_i b_i + a_j b_j.$$

The claim is symmetric in i and j , so assume $j \leq i$. Both sequences are increasing, so $a_j \leq a_i$ and $b_j \leq b_i$; write $a_i = a_j + u$ and $b_i = b_j + v$ with $u, v \in \mathbb{N}$. Then

$$a_i b_j + a_j b_j = 2a_j b_j + a_j v + u b_j + uv, \quad a_i b_j + a_j b_i = 2a_j b_j + u b_j + a_j v,$$

so the first exceeds the second by exactly $uv \geq 0$. This is the only point in the argument at which the hypothesis is used, and what it supplies is that $a_j \leq a_i$ and $b_j \leq b_i$ hold simultaneously.

Summation. We use the fact that an inequality may be summed: if $x_{ij} \geq y_{ij}$ for all i, j then $\sum_{i,j} x_{ij} \geq \sum_{i,j} y_{ij}$. This follows from the two-term case $x \geq y$ and $x' \geq y'$ imply $x + x' \geq y + y'$, by induction on the number of terms. It is a fact of a different kind from the identities used below to rearrange sums, and is worth stating separately.

Summing the lemma over all i and all j , the left side becomes

$$\sum_i \sum_j (a_i b_j + a_j b_i) = \sum_i \left(\sum_j a_i b_j \right) + \sum_i \left(\sum_j a_j b_i \right) = n \sum_i a_i b_i + n \sum_j a_j b_j = 2n \sum_i a_i b_i,$$

where the inner sum in the first term has n equal summands and the inner sum in the second term does not depend on i . The right side becomes

$$\sum_i \sum_j (a_i b_j + a_j b_i) = \left(\sum_i a_i \right) \left(\sum_j b_j \right) + \left(\sum_j a_j \right) \left(\sum_i b_i \right) = 2 \left(\sum_i a_i \right) \left(\sum_i b_i \right),$$

by distributivity. Hence $2n \sum a_i b_i \geq 2 \left(\sum a_i \right) \left(\sum b_i \right)$, and cancelling the factor 2 – legitimate because $2x \geq 2y$ implies $x \geq y$ in $\mathbb{N}$ – gives the result. $\square$

Remark. No induction on n was needed beyond the standard facts about finite sums. Note also that the proof consumes less than is assumed: it never needs either sequence to be sorted, only that the two agree on which of any two indices carries the larger entry. If instead $a_1 \leq \dots \leq a_n$ and $b_1 \geq \dots \geq b_n$, the same pairwise step with the two sides exchanged reverses the inequality.

Rubric (8 Marks)

1. Stating and deriving the pairwise lemma

- Correct derivation of $a_i b_j + a_j b_i \geq a_i b_i + a_j b_j$ through any kind of manipulation of inequalities or by the use of algebra – 3M
- A use of “–” sign could be penalised by a 0.5M over the total score for this section of the solution

2. Some statement over summation of the inequalities as series

- Requires the explicit mention of something along the lines: if $x_{ij} \geq y_{ij}$ for all i, j then $\sum_{i,j} x_{ij} \geq \sum_{i,j} y_{ij}$, with at least some direction of the proof as to why (two-term case plus induction on the number of terms) – 2M

- Partial credit of 1M if the sum is used correctly but treats it as obvious/unstated.

3. Correctness of the working related to it

- LHS collapses to $2n \sum a_i b_i$, with the counting of n repeated terms – 1M
- RHS collapses to $2(\sum a_i)(\sum b_i)$ via distributivity – 1M
- Final cancellation of the factor of 2 (legitimate in $\mathbb{N}$) and a clean conclusion – 1M

Question 5 Rubric

Real Analysis — Assignment 1

Grading Points

- **Derive the translation property – 2 marks:** Starting from $f(f(n)) = n++$, apply f to both sides and compare with the original equation applied at $f(n)$ to obtain

$$f(n++) = f(n)++.$$

- **Express $f(n)$ in terms of $f(0)$ – 2 marks:** Using the translation property repeatedly or by induction, show that

$$f(n) = f(0) + n$$

for every $n \in \mathbb{N}$.

- **Obtain a contradiction and conclude non-existence – 4 marks:** Let $c = f(0)$. Substitute $f(n) = c + n$ into $f(f(n)) = n++$ to obtain

$$n++ = c + c + n,$$

which implies

$$c + c = 0++,$$

which is impossible for $c \in \mathbb{N}$. Hence, conclude that no function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfies $f(f(n)) = n++$ for every $n \in \mathbb{N}$.

Any other mathematically correct method that obtains a contradiction and establishes non-existence will receive full marks for this step.

- **Suggest a function for the second part – 1 mark:** Identify/suggest the function

$$f(n) = n++.$$

Note: Any other correct function will fetch full marks for this step.

- **Verify the function – 1 mark:** Verify that

$$f(f(n)) = f(n++) = n++++.$$

Hence, the proposed function satisfies the required condition.

General Notes

- **Any fully correct mathematical solution will receive full marks.** Alternative approaches are acceptable as long as the reasoning is logically valid and completely establishes the required result.
- For the contradiction step, any correct argument proving that such a function cannot exist will receive the full 3 marks allocated to that step.
- Minor differences in notation, such as writing $n + 1$ instead of $n++$ or $n + 4$ instead of $n++++$, should not be penalized if the mathematical meaning is clear and correct.
- For the second part, the student must provide a valid function and verify that its composition with itself gives the required expression.

Grading Rubric: Question 6 [8 Marks]

MA4.101 Real Analysis (Monsoon 2026) — Assignment 1

Prove that $\{X, \mathcal{P}(X), \mathcal{P}(\mathcal{P}(X)), \dots\}$ is a set.

What the question is testing. The collection is not given; it must be constructed. The only axiom that can produce it is replacement (Axiom 6) applied to $\mathbb{N}$, and replacement needs a property $P(n, Y)$ that determines Y uniquely from n . Writing “ $Y = \mathcal{P}^n(X)$ ” presupposes that $n \mapsto \mathcal{P}^n(X)$ is already defined, which is the substance of the problem. Marks are therefore weighted towards making the rule precise.

Marking scheme

	Criterion	Marks
A	Framing. Recognises that the ellipsis must be given a meaning and that the set cannot be assumed. Notes (briefly) that specification needs an ambient set that is not available, and that the recursion theorem needs a fixed target set, so neither applies directly.	1
B	Precise rule. Gives an explicit first-order definition that pins down $\mathcal{P}^n(X)$: e.g. n -towers (graphs of functions on $\{k \leq n\}$ with $s(0) = X$ and $s(k++) = \mathcal{P}(s(k))$), or any equivalent “finite approximation” device.	1.5
C	Existence of approximations. Induction on n : base $\{\langle 0, X \rangle\}$ via singleton; step $s_n \cup \{\langle n++, \mathcal{P}(s_n(n)) \rangle\}$ via power set, singleton, pairwise union; verifies the defining clauses.	1.5
D	Uniqueness of approximations. Shows any two n -towers coincide (via restriction to $\leq n$ plus induction, or a direct induction on $k \leq n$ showing $s(k) = t(k)$).	1.5
E	Correctness of the rule. Shows $\mathcal{P}^0(X) = X$ and $\mathcal{P}^{n++}(X) = \mathcal{P}(\mathcal{P}^n(X))$, i.e. the defined objects are the intended ones.	0.5
F	Replacement. States a property $P(n, Y)$ in terms of the construction (e.g. “some n -tower contains $\langle n, Y \rangle$ ”), verifies that for each $n \in \mathbb{N}$ exactly one Y satisfies it (uses C and D), and applies Axiom 6 with $A = \mathbb{N}$ (a set by the axiom of infinity).	1.5
G	Conclusion. Identifies the resulting set $\{\mathcal{P}^n(X) : n \in \mathbb{N}\}$ with the collection in the statement.	0.5
	Total	8

Caps for structurally flawed answers

Apply the lowest applicable cap, then award within it using the table above.

Answer type	Max
Names the collection T up front and applies specification or “it is a subset of $\dots$ ” to conclude it is a set (circular).	1
Applies replacement to $\mathbb{N}$ with $P(n, Y): Y = \mathcal{P}^n(X)$, where $\mathcal{P}^n$ is defined only informally (“apply $\mathcal{P}$ n times”) or by an unjustified recursion. Award F and G if the replacement step itself is stated correctly.	3
Invokes Tao’s recursion theorem (Prop. 2.1.16) or a recursion into “the set of all sets” to define $n \mapsto \mathcal{P}^n(X)$. Recursion theorem in Tao produces sequences in $\mathbb{N}$; a “set of all sets” does not exist.	3
Correctly cites a general set-theoretic recursion theorem (e.g. ZF recursion on ω with a class function) without proof, then applies replacement properly.	5
Full tower-style construction but uniqueness (D) omitted, so the property in F is not shown to be functional.	5.5

Partial credit and deductions within criteria

- **B:** Definition that fixes values but not the domain (e.g. omits “exactly one Y for each $k \leq n$ ”), or omits $s(0) = X$: award 1 of 1.5.
- **C:** Construction correct but no axiom citation for why s_{n++} is a set: -0.5 . Clause (ii) (no duplicate pair at $n++$) not checked: -0.5 .
- **D:** Uniqueness argued by “clearly the values are forced” with no induction: award at most 0.5 of 1.5.
- **E:** May be merged into F or G; award if the content is present anywhere.
- **F:** Replacement applied without checking uniqueness of Y : -1 . Domain of replacement not stated to be a set: -0.5 .
- **Minor slips** (e.g. $k < n$ vs. $k \leq n$, off-by-one in indexing) that do not break the argument: -0.5 total, not per occurrence.

Alternative approaches (full credit if complete)

- **Functions on initial segments.** Define an n -approximation as a function $g : \{0, \dots, n\} \rightarrow Y$ into some set Y with $g(0) = X$, $g(k++) = \mathcal{P}(g(k))$. Care is needed: the codomain must exist; students must either avoid fixing a codomain (graph-only, as in the solution) or construct one by induction. Map criteria B–F one-to-one.
- **Intersection of inductive-type sets.** Not directly available (there is no ambient set in which to intersect); treat as the circular case unless an ambient set is genuinely constructed first.

Q7 Rubric

Total Marks: 6

1M: Constructing D and showing that the proposition

$$P(x) : x \notin f(x)$$

is a valid proposition.

1.5M: Using the Axiom of Specification on X to obtain

$$D = \{x \in X \mid x \notin f(x)\},$$

and concluding that

$$D \subseteq X,$$

hence

$$D \in \mathcal{P}(X).$$

1.5M: Contradiction approach: assume f is surjective. Then there exists

$$a \in X$$

such that

$$D = f(a).$$

1M: Deriving

$$a \in D \iff a \notin f(a) \iff a \notin D.$$

1M: By contradiction, no such a can exist. Therefore D is not in the image of f , so f is not surjective.

Assignment Q8 Rubric

Total Marks: 10

- **Part A:**

- 1 mark for writing the domain and range match for both the LHS and RHS.
- 1 mark for taking an arbitrary x and proving the required equality.

- **Part B:**

- 1 mark for writing the domain and range match for both the LHS and RHS.
- 0.5 marks for proving the first equality.
- 0.5 marks for proving the second equality.

- **Part C:**

- 1 mark for proving that f^{-1} exists.
- 1 mark for writing the domain and range match for both the LHS and RHS for both equalities.
- 1 mark for taking an arbitrary x and proving both equalities.

- **Part D:**

- 1 mark for defining the piecewise function and showing that it is a valid function.
- 1 mark for proving the required identities in the question.
- 1 mark for showing its uniqueness.